

What Term Limits Move: Effort, Positions, and the Dynamics of State Legislatures

Kisoo Kim* Craig Volden[†] Alan Wiseman[‡]

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Abstract

Legislative term limits present two puzzles that mirror each other. Term-limited chambers polarize, yet individual positions barely move; termed-out legislators disengage, yet aggregate lawmaking does not decline. We resolve both with difference-in-differences against never-limited chambers and within-legislator comparisons, using bill histories in 49 states and a new 34-chamber roll-call panel. Chambers polarize because limits expel veterans recruited in a more moderate era, so the chamber comes to reflect its more polarized recent entrants, not because any member moves; once that generation is gone, the added polarization shows no detectable growth, and entry, exit, and within-career change match ordinary elections. Lawmaking is unchanged because the disengagement is selective: termed-out legislators drop the long-shot bills, about 7 percent of their sponsorship, while their enacted law is untouched, freshmen close two-thirds of the newcomer sponsorship deficit, and women pass more law within their own careers. Term limits restructure the chamber's ideological composition and redistribute its legislative work.

*Email: kisoo@virginia.edu

[†]Email: volden@virginia.edu

[‡]Email: alan.wiseman@vanderbilt.edu

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1 Introduction

Legislatures that adopted term limits pulled apart. Relative to legislatures that never adopted them, the distance between the two parties in adopting chambers widened once the limits bound (Olson and Rogowski, 2020). Yet at the individual level almost nothing moves. Termed-out members hold their positions essentially fixed (Fouirnaies and Hall, 2022), and legislators' revealed ideology is among the most stable quantities in American politics (Poole and Rosenthal, 2007). A chamber whose members do not move has nonetheless moved.

The same inversion appears, reversed, on the effort dimension. A legislator entering her final permitted term under a term limit has lost the incentive that structures most of legislative life: the next election. The behavioral consequences are well documented. The same termed-out legislators whose positions stand still sponsor fewer bills, perform less committee service, and are present for fewer floor votes (Fouirnaies and Hall, 2022). The literature reads that pattern as shirking, one entry in a long ledger of term limits' institutional costs alongside lost expertise, weakened chambers, and empowered lobbyists (Carey et al., 2000, 2006; Kousser, 2005).

Yet the legislatures these final-term members populate do not produce less law. California's Assembly passed bills at essentially the same rate while Proposition 130 cycled termed-out members through it (Cain and Kousser, 2004), and in the first states where limits bound, what changed was less the volume of lawmaking than its distribution: legislative success migrated from minority to majority members (Kousser, 2005). If a substantial share of every term-limited chamber shirks each term, the arithmetic of aggregate stability should not work. Fouirnaies and Hall (2022) recognize exactly this tension: their own chamber-wide extrapolation (63 fewer sponsored bills per term) explicitly sets aside "possible general-equilibrium effects of term limits." So each dimension contradicts itself across levels, and the two dimensions contradict each other: positions move in the aggregate but are stable in the individual, while output is stable in the aggregate but moving

in the individual.

This paper asks what term limits actually change in a legislature, and answers with a single distinction. A chamber's ideological location is embodied in its people. It changes only as fast as its people change, and term limits change the people once: when the limits first bind, they expel the veterans accumulated over decades when entrants of both parties stood nearer the center. After that one-time compositional shift, entry and exit under the limit are ideologically indistinguishable from ordinary electoral turnover. Effort, by contrast, is chosen anew every term under whatever incentives the institution supplies. Term limits alter those incentives for the member whose horizon just ended and for the newcomer whose expected tenure is now shorter from the day of arrival, so the effort reallocation recurs in every cohort thereafter.

The position findings and the effort findings discipline each other. If the polarization effect were not a one-time compositional shift but an ongoing engine (extremist entrants, moderate-removing exits, members radicalizing in place), it should compound rather than plateau, and it should be visible at the margins we can observe member by member. Instead, the effect appears at implementation, shows no detectable growth beyond the first decade (the time 6–12-year caps need to rotate out the inherited membership), and every member-level channel that could sustain it is closed thereafter. If the effort effects were compositional (different people rather than changed behavior), they should appear between legislators, not within them. Instead, the effects that admit a within-person design (the final-term withdrawal, the female gain) survive it, and all of them recur in every era of the reform.

For positions, we build a 34-chamber roll-call panel (22 term-limited chambers and 12 never-limited control chambers, scored per term on an identical pipeline) covering roughly 33,000 legislator-terms from 1991 to 2025, supplemented by bridged ideal-point estimators that make within-person movement comparable across time and by the Shor and McCarty (2011) common space for cross-state levels. For effort, we trace individual

legislators' bills through each stage of the legislative process from 1993 to 2017, using the bill histories underlying the State Legislative Effectiveness Scores (Bucchianeri et al., 2025). The treated states of the two panels overlap almost completely, and we show that each panel's findings hold on the intersection: the position findings hold within the years where we measure effort, and the effort findings are unchanged when restricted to the states where we measure positions.

On the position side, we confirm the chamber-level polarization effect documented by Olson and Rogowski (2020), but it is a transition effect: it emerges as the limits expel the accumulated moderate veterans, a one-time level shift added to the outward drift every legislature shares, and that added polarization shows no detectable growth once the expulsion is complete. In the steady state that follows, the limit adds no polarization beyond what ordinary electoral turnover already produces: entrants are no more extreme than the incumbents they join; the limit's exits, made on schedule rather than on ideology, remove no more moderates than elections would have; and members' positions change within their careers no more under the limit than without it. On the effort side, we find that the final-term decline is selective rather than general: termed-out legislators abandon the long-shot bills that would have died in committee, while their legislation that becomes law is untouched. The shed effort is compensated: freshmen in term-limited chambers close most of the sponsorship deficit that normally separates newcomers from veterans, and women pass more law within their own careers under the limit. Aggregate output is preserved while the distribution of legislative success shifts toward juniors.

These findings contribute to three conversations. First, they estimate the time path of the term-limits polarization effect (Olson and Rogowski, 2020) and locate it at a single margin, the one-time expulsion of the chamber's accumulated moderate veterans, which reverses the reform-design lesson: the polarization cost of term limits is a fixed transition cost, not a running one. Second, they resolve the aggregate-effort question Fourinaies and Hall (2022) left open and reconcile the shirking literature with the aggregate-stability

findings that have accumulated since the first post-implementation studies (Cain and Kousser, 2004; Kousser, 2005): individual decline concentrated in bills that would not have passed, plus compensation from newcomers, equals stable output. Third, they offer a template for evaluating institutional reforms generally. A reform changes an institution through composition (who sits in it) and through incentives (what sitting members choose); the first effect is spent once the membership turns over, while the second recurs as long as the rules stand. Conflating the two produces exactly the micro–macro contradictions this literature has carried: a one-time compositional shift misread as an ongoing engine, and a lasting incentive change misread as aggregate decline.

2 Two Clocks: How Term Limits Change a Legislature

Legislative behavior is organized by the electoral connection (Mayhew, 2004). Members' voting records express commitments, to districts, donors, and selves, that are notoriously durable (Poole and Rosenthal, 2007), and members allocate scarce time and staff across activities with an eye to reelection. Term limits sever the electoral connection at a known date. What follows from that severance depends on which object one tracks: the chamber's ideological location is embodied in its members, so it changes only as fast as the membership turns over; effort is rechosen every term, so it responds to the new incentives immediately and in every cohort thereafter. The two objects run on different clocks.

2.1 Positions: Embodied in Persons

A chamber's ideological location can change through exactly three channels: who arrives (entry), who leaves (exit), and who changes in place (conversion). Term limits act directly on exit alone: the reform adds a second exit door beside the electoral one, so that a chamber now loses members through two routes. Elections remove members selectively, when challengers beat them; the mechanical door removes members on schedule, regardless

of ideology. Everything the reform does to a chamber's positions must therefore operate through the members the mechanical door removes (exit), through the entrants recruited to the seats it opens (entry), or through sitting members changing their positions in response to the limit (conversion).

It is critical to make a distinction between the limit's first binding and its steady-state operation. What made the veteran membership ideologically distinctive was not how it survived but when it was recruited. State legislatures have polarized for a generation (Shor and McCarty, 2011), and the polarization rose through entry of new members: each cohort of entrants stands a little further from the center than the last. A sitting chamber is therefore a layered archive of its own recruitment history, juniors embodying the current entry distribution and veterans the distributions of one, two, and three decades ago; its ideological location is a duration-weighted average of past entry cohorts, a memory of its own recruitment. Every veteran is, by definition, an electoral survivor, and because elections remove the relatively moderate faster, as we show below, the survivors of a cohort are on average its less moderate members. But veterans were also recruited in a more moderate era, and that gap across cohorts is larger: even the selected survivors sit nearer the center than their juniors. What seniority mainly marks, ideologically, is vintage (Figure 1).

A term limit truncates that average's window: no sitting member can have entered more than a cap length ago, so the chamber's location comes to reflect only its most recent entry cohorts. When limits first bind, they expel the oldest vintages at once, a one-time release that should shift the chamber's location over exactly the horizon the caps prescribe: six to twelve years, the time for the inherited membership to rotate out. After that, the limit removes each cohort as it reaches the cap, and the average runs over a constant, shorter window.

That release changes who sits in the chamber, and it is complete once the accumulated vintages are gone. Whether the limit polarizes the chamber beyond that mechanical com-

positional shift, in the steady state and relative to control chambers, depends on the three channels: it does only if the term limits remove moderates at a higher rate than elections would have (exit), if the seats it opens recruit more extreme entrants (entry), or if serving under the limit radicalizes members in place (conversion). Each channel is observable member by member against control chambers. If all three are absent, the polarization effect is a transition effect (a one-time replacement of members, not a change in the ongoing turnover), and it should build, plateau, and not compound.

2.2 Effort: Rechosen Every Term

Lawmaking effort is a portfolio rather than a single quantity. Introducing a bill costs little more than drafting time; moving one costs sustained work at every subsequent stage: securing a hearing, negotiating amendments, assembling majorities. A member who loses the reelection incentive can reduce total effort in two ways: do less of everything, or concentrate the costly late-stage work on fewer, better bills. The first is shirking, and the second is selectivity. The two make identical predictions about sponsorship counts but opposite predictions about the pipeline: shirking predicts a decline that persists or compounds toward enactment, where effort costs are highest; selectivity predicts a decline that attenuates stage by stage and vanishes for bills that become law, because the motives that survive the final term (policy goals, reputation, legacy) attach to statutes, not introductions.

Term limits change the incentives not only of members in their final term but also of everyone else in the chamber, and most sharply of its juniors; we call this the horizon mechanism. In an unlimited chamber a freshman expects a long time horizon: seniority norms, committee queues, and the option value of waiting all counsel patience. A term limit collapses that option value on the day it is enacted. A first-term member who will be barred in eight years has no future in which deferred effort pays, so juniors should legislate like full members from the start. Two features of this argument matter for testing it. It oper-

ates through expectations, not through resources: it does not require that juniors inherit the departing seniors' committee posts or agenda space, and so it predicts a *diffuse* junior response rather than one concentrated in post-holders. And it operates at the level of the regime, not the dose: the boost should appear wherever a binding cap exists, should not scale with how many seniors happen to be exiting in a given term, and should already be visible after enactment but before the limit expels anyone. The alternatives are compensation flowing through vacated posts, through the collapse of apprenticeship norms as veteran enforcers exit, or through the selection of differently ambitious entrants. The first two are triggered by exit, and each implies a signature the data can check: gains concentrated in the members who inherit the vacated posts, or a boost that appears only once veterans actually leave and grows with how many leave. The third is not fully separable from the horizon mechanism itself, because a cap changes who runs from the day it is enacted: entrants selected under the cap would show the same immediate, regime-level, dose-flat profile as ordinary entrants whose behavior the shortened horizon changes. We test the two exit-triggered alternatives directly and state below which version of the selection alternative the evidence excludes.

The composition–incentive distinction also explains why the two dimensions of the same reform behave so differently. Positions respond once because they are embodied in persons; once the inherited membership has rotated out, the door's turnover matches what elections already produced. Effort responds continuously because it is rechosen every term; each new cohort of termed-out members re-enacts the withdrawal, and each new cohort of freshmen re-enacts the surge. The observable signature of the contrast is temporal: a rise that is complete within the transition decade and flat thereafter on the position side, sustained level shifts in behavior on the effort side.

3 Data and Design

3.1 Two Panels

The analysis of positions uses a roll-call panel we constructed from the states' own voting records: 34 chambers (22 term-limited and 12 never-limited controls) scored with per-term W-NOMINATE (Poole and Rosenthal, 1985) on an identical pipeline, roughly 33,000 legislator-terms from 1991 to 2025. The construction follows and extends Fourniaies and Hall (2022), who scaled the roll calls of term-limited chambers term by term from the states' own records and compared each member with herself across terms; their scaled sample covers nine states through 2016. The panel here scales all nine and three more term-limited states, both chambers in ten of the twelve, runs the series to 2025, and adds both chambers of six never-limited states as controls, which a within-legislator design did not need but the regime-level contrasts below require. The score places each member on a single ideological scale running from -1 to $+1$, oriented so that Republicans take positive values, with zero at its center; a member's own-party pole is the end of that scale on her party's side. Chamber-by-term fixed effects absorb each chamber-term's own level, so the identifying variation is always the member-versus-member comparison inside a chamber-term, never a cross-scale level difference. Because per-term scaling can manufacture spurious within-person movement, every within-person claim is adjudicated on two estimators whose failures run in opposite directions. Per-term W-NOMINATE rescales every term independently, so a term-specific stretch of the score space can register as movement of every member; a fixed-item estimator in the spirit of Nokken and Poole (2004) pins the issue space across a chamber's terms from a pooled scaling in which each member holds one position, so a movement common to a whole party can be absorbed into the items. Both measure a member's position relative to the chamber-term, and a within-person finding counts only if the two estimators agree in sign and each is distinguishable from zero at the 5 percent level (the two-estimator rule). A bridged dynamic

item-response (IRT) model (Imai et al., 2016), which links each member's positions across terms through a random-walk prior while estimating the items term by term, is reported beside them but does not adjudicate: nothing but that prior ties the scale of one term to the next, and on this panel its scale widens term after term, which the member-level contrast reads as movement of continuing members (Appendix A). Cross-state *levels*, the size of the adoption effect, come from the Shor and McCarty (2011) common space (April 2023 release); because the fixed-item estimator pins the issue space within a chamber across its terms, the distance between a chamber's two party medians can also be followed over time on the roll-call panel itself, though not compared across chambers.

The analysis of effort uses the bill-history data underlying the State Legislative Effectiveness Scores (SLES; Bucchianeri et al., 2025): for every legislator-term in 47 states from 1993 to 2017, the number of bills reaching each of five stages (introduction, committee action, action beyond committee, chamber passage, and law), separately for commemorative, substantive, and substantively significant legislation. The full panel contains 76,303 legislator-terms. Three states are absent. Kansas lacks the sponsorship data the scores require. The member-level panel is restricted to states that nominate legislators through party primaries; Nebraska and Louisiana, the two other states that operated term limits in the period, fail that restriction and are excluded: Nebraska's unicameral legislature is nonpartisan, so party and majority status are undefined for its members, and its bill histories enter the scores only in 2007, after its limits had begun removing members; Louisiana's blanket primary places every candidate on one ballot regardless of party. Chamber-level aggregates draw on a chamber-term panel built from the same bill histories for 49 states; the chamber-level estimates use the chamber-terms with complete chamber covariates, 43 states, and their thirteen treated states are the thirteen of the member-level panel. Bill counts by stage measure absolute output. The SLES combines those fifteen stage-by-type counts into a single weighted score (commemorative bills weighted 1, substantive 5, substantively significant 10), normalized within each chamber-term so

that the average member scores 1; computing the same weighted sum without the normalization gives the unnormalized SLES, in absolute units. We use the bill counts and the unnormalized SLES to test whether output changes, and the SLES to test whether its distribution changes.

Thirteen of the effort panel's states operated binding limits during the study period, adopted in the early-1990s wave and binding from 1996 onward; following Butcher (2023) we distinguish lifetime limits (Arkansas, California, Michigan, Missouri, Nevada, Oklahoma) from consecutive-service limits (Arizona, Colorado, Florida, Maine, Montana, Ohio, South Dakota). The two panels cover the same institutions: twelve of the thirteen treated states in the effort panel appear in the roll-call panel (Colorado's roll calls are collected, but its member records carry no party identifiers, so its chambers are not yet in the panel), and Appendix D shows every position headline holds inside the effort panel's 1993–2017 window and every effort headline is unchanged when the treated set is restricted to the overlap. Summary statistics for both panels appear in Appendix Table B1. An entrant in the roll-call panel and a freshman in the effort panel are the same object, a member in her first term.

3.2 Designs

Term limits generate treatment variation at two levels. At the regime level, a chamber operates under binding limits and can be compared with chambers that never adopted them; at the individual level, a member reaches her final permitted term and can be compared with herself before it. The levels differ in what they can identify. When a member reaches her final term follows mechanically from her entry date and the cap length, so a contrast of a member with herself before her final term does not depend on when or why her state adopted limits. It does depend on who serves under them: limits may change who enters and who stays, and the final-term effect it recovers is the effect among the members the regime admits and retains. Composition is itself one channel of the regime's effect, which

the entry and exit contrasts estimate. Every contrast that compares term-limited with never-limited chambers (the regime-level designs, and the member-level entry, exit, and drift differentials alike) rests on the comparability of adopting and non-adopting states, and adoption is not random. The designs below each estimate one of the two treatments; the unit of analysis for every member-level estimate, on both panels, is the legislator-term, one member observed in one two-year legislature.

On the position side, the regime's effect on the chamber's own location is estimated twice, on chamber-level outcomes; every member-level design absorbs that quantity in its chamber-by-term fixed effects. Its size, and the comparison across states, come from the Shor and McCarty (2011) common-space scores: we replicate the Olson and Rogowski (2020) difference-in-differences against non-adopting states and add its time path, with a heterogeneity-robust event-study estimator (Sun and Abraham, 2021). Its time path is then estimated again on the roll-call panel's own fixed-item scores, on which the distance between a chamber's party medians is comparable across that chamber's terms: the outcome is the log of that distance in each chamber-term, the specification is the same event-time design with chamber and biennium fixed effects and the never-limited chambers as controls, and the panel runs to 2025 rather than 2016. Because those scores carry no common metric across chambers, the second estimate speaks to the sign and shape of the effect, not to its size. Together the two event studies are the pre/post evidence on positions.

Member-level position contrasts take each roll-call score in its own units as the outcome, the measure of Fourniaies and Hall (2022), which we call extremity: the absolute score when the parties are pooled, the score signed toward the own-party pole within party (a higher value is closer to -1 for a Democrat and to $+1$ for a Republican), and the signed score for the gap between the parties, with the bridged scores rescaled to unit variance within each chamber across all of its terms; the never-limited chambers are the benchmark throughout. Within-person contrasts are estimated pooled across chambers

with person and chamber-by-term fixed effects, and chamber by chamber as a census: the termed-out shift, per party; career drift as a member’s slope in terms served, per party, with its term-limited differential; and the Republican-minus-Democratic differential in that slope with its term-limited differential. Entry and exit contrasts compare entrants (or exiters) with stayers: the estimate is the extremity gap between the two groups in term-limited chambers minus the same gap in never-limited ones, with chamber-by-term fixed effects, which absorb the regime main effect, and clustering by chamber. Following Fourinaies and Hall (2022), the member-level contrasts exclude chambers whose members serve staggered four-year terms, in which a member’s final permitted term spans two biennia and the two-year term that follows each redistricting makes the final-term coding approximate: the eight such term-limited senates and the two such never-limited senates (Alaska and New Jersey) leave, and the member-level sample is 24 chambers, the twelve term-limited lower chambers with the two-year-term senates of Maine and South Dakota against the six never-limited lower chambers with the two-year-term senates of Connecticut, North Carolina, New Hampshire, and Vermont. Estimated on all 34 chambers instead, every verdict in Tables 2 and 3 is the same: the pooled final-term shift is +0.000, −0.010, and −0.008 on the three estimators, and the entry and exit differentials are −0.012 and +0.008 against the −0.016 and +0.006 reported below. Because five of the 14 term-limited chambers enter our panel only after their limits had bound, these pooled member-level contrasts are steady-state comparisons against the never-limited controls, not pre/post designs.

On the effort side, the same two treatments are estimated on the bill-history panel. The within-career design estimates the individual treatment, the final-term response, in the thirteen term-limit-binding lower chambers:

$$y_{ict} = \beta \text{Final}_{it} + X'_{it}\gamma + \alpha_i + \delta_{ct} + \varepsilon_{ict}, \quad (1)$$

where y_{ict} is an output measure for legislator i in chamber c and term t , Final_{it} marks her final permitted term, X_{it} collects individual controls, and the legislator fixed effect α_i and chamber-by-term fixed effect δ_{ct} confine the comparison to the same member across terms, net of everything common to her chamber in a term, including the regime itself: δ_{ct} absorbs the regime indicator, which therefore cannot enter equation (1). β is identified from within-career change among the 1,325 members observed both before and in their final term; it does not depend on adoption timing, and the coefficient on the term before the final permitted term is +0.7 bills sponsored (standard error 0.4), positive and not distinguishable from zero, so no anticipatory decline above about 0.8 bills, smaller than the final-term decline itself, is present.

The cross-member design estimates the regime treatment on the same unit:

$$y_{ict} = \beta \text{TL}_{ct} + X'_{ict}\gamma + \phi_c + \psi_t + \varepsilon_{ict}, \quad (2)$$

where TL_{ct} equals one when chamber c operates binding limits in term t , and the chamber and term fixed effects enter separately so that the regime indicator is not absorbed. Estimated on all lower chambers, equation (2) asks whether output differs at all in chambers operating under binding limits. Final-term members remain in the sample deliberately: the withdrawal that equation (1) isolates enters β here, together with every response to it, so β is the general-equilibrium net effect of the regime.

The theory predicts responses that differ within the chamber. The collapse of the option value of waiting should move juniors most, and the gender literature locates a second distinctive response among women (Holman and Mahoney, 2023). We therefore extend (2) with a group indicator G_{ict} and its interaction with the regime, estimating the extended equation twice, once with G_{ict} marking freshmen and once with it marking women:

$$y_{ict} = \beta (\text{TL}_{ct} \times G_{ict}) + \lambda_1 \text{TL}_{ct} + \lambda_2 G_{ict} + X'_{ict}\gamma + \phi_c + \psi_t + \varepsilon_{ict}. \quad (3)$$

The interaction coefficient β measures whether the group’s output differs in chambers operating under binding limits, over and above the group’s baseline gap (λ_2) and the regime’s overall level (λ_1). Equation (3) is estimated on a sample that excludes final-term observations outright, so it measures the regime’s effect apart from the withdrawal of the termed-out legislators, and the group contrasts are not inflated by it. Standard errors are clustered by state in both designs. Because adoption is not random, we read cross-member estimates as conditional associations and base the causal claims on the within-career design; where both designs can answer the same question, their agreement is itself evidence.

The chamber-level effort outcomes, the four distribution measures of Appendix Table B4, are estimated in the same two-way form on the chamber-term panel: each measure on the regime indicator with state-chamber and year fixed effects, conditioning on the concurrent reforms (redistricting commissions, primary openness, the top-two primary, professionalism) and the chamber controls (polarization, seat-share imbalance, log chamber size) that the table note lists, with standard errors clustered by state-chamber.

The member-level position contrasts are these forms with a different outcome: the termed-out shift is equation (1) with extremity as y , pooled across the term-limited chambers and chamber by chamber, and the entry and exit differentials are equation (3) with G marking entrants or exiters and chamber-by-term fixed effects in place of the separate chamber and term effects.

4 Positions

4.1 The Polarization Effect Is Real, and It Is a Transition

We replicate Olson and Rogowski (2020): on Shor and McCarty (2011) scores, lower chambers that adopted term limits polarized relative to non-adopters. In the simplest specification, the outcome is the distance between the two parties’ median scores in each chamber

and year; the treatment is a chamber-level indicator equal to one from the first year the chamber's limits bind onward; and with state and year fixed effects, the estimate averages all of those years into a single number. In term-limited chambers, that distance is 0.128 common-space units larger (standard error 0.047), roughly 9 percent of the 1.40 average distance between the party medians in never-limited chambers over the same years, and the heterogeneity-robust estimator of Sun and Abraham (2021) returns 0.135.

That single number, however, cannot say when the gap emerged or how it evolved. Did the gap appear as soon as the limits were enacted, years before anyone had been expelled, or only once the caps began forcing seniors out? And did it keep growing as the membership turned over, or arrive at full size at once? To ask them, we put each adopting chamber on its own clock. Year zero is the chamber's implementation year, the first year the limits bind and its first full year without the members the cap has just removed; the election that removed them falls at the end of year -1 , so that cohort's final term in office lies entirely in the lead years. We estimate the polarization gap separately over six windows of that clock: years -7 and earlier and years -6 to -4 before implementation, and years 0 to 3 , 4 to 7 , 8 to 11 , and 12 onward after it. Each coefficient measures how much larger the party-median distance was in adopting chambers than in never-limited ones during those clock years, relative to the three years just before implementation (-3 to -1). The six coefficients, read in order, are the gap's time path, and they add three facts (Table 1, Panel A).

First, the gap appears early. In the first four years after the first termed-out cohort was removed, most seats were still held by members who had entered before the limits, yet the estimate for those years is already distinct from zero under both estimators, and the point estimates stay positive in every later window through two decades, the last window being the least precise. Second, the point estimates are larger in the later windows and flat after the first decade, and the span over which they rise matches the 6–12 years the caps need to force out the last members who entered before the limits. But

the four post-implementation windows cannot be distinguished from one another, so an effect that grew as those members were replaced and an effect at full size from the start fit the estimates equally well. Third, the pre-implementation estimates are indistinguishable from zero, individually and jointly, and the nearer lead is precise enough to matter: it would detect a gap above about 0.06 common-space units, smaller than the first post-implementation window's 0.08, while the far lead, resting on one to three states, detects only gaps above about 0.10. The years before -6 , which for nearly every adopting state precede the law's passage, carry the same gap as the three years just before the first expulsion, which follow it. The law's passage alone produced no detectable movement in positions; the gap dates from the expulsions.

The same event-time design can be run on the roll-call panel itself (Table 1, Panel B). On the fixed-item scores the distance between a chamber's party medians is comparable across that chamber's terms, so we estimate the six windows again on the log of that distance for all 34 chambers from 1991 to 2025, with chamber and biennium fixed effects and the never-limited chambers as controls. The first two facts reappear, the second in a stronger form: the gap is larger in term-limited chambers from the first post-implementation window on, and thereafter it neither rises across the first decade nor changes after year twelve: the four post windows are jointly distinct from zero and indistinguishable from one another. Because these scores run nine years past the common space, the path can be followed further, and it stays flat: the estimate for years 20 and beyond equals the estimate for years 12–19, and the estimate for the chamber-terms after 2016 does not differ from that for the years the common space covers. The third fact returns with a qualification. Relative to the biennium just before implementation, the far lead (years -7 and earlier) is lower on the roll-call scores, where the common-space leads were flat; the estimate is not distinguishable from zero and it shrinks when the minimum of scored members per party is raised from five to eight, but it is identified from the three chambers observed seven or more years before implementation, so an onset one biennium

earlier than the common space dates it cannot be excluded on this panel; either anticipatory departures ahead of the cap or anticipatory voting would produce it. Roll-call scores carry no common metric across chambers, so this reading speaks to sign and shape, not to size; a vote-direction measure that needs no scaling model reproduces the plateau but not the increase at implementation (Appendix A).

One question the time path cannot answer is whether the process is finished. If polarization grew because the members who entered before the limits were gradually replaced, it should stop once that replacement is complete; the flat estimates after year twelve, which on the roll-call panel run to 2025, fit that reading. But an effect still growing too slowly to detect over nearly three decades would produce the same flat estimates, so the time path cannot reject that reading either. The member-level tests that follow distinguish the two.

Every member-level channel that could keep the effect running (entry, conversion, exit) is closed in the steady state, and the one margin at which anything ever changed is the door's first opening.

One accounting note governs what follows. The roll-call panel shows the transition on its own scores (Table 1, Panel B), so the channel tests below examine the phenomenon in the data where it is observed; but they are denominated in the roll-call scores' own units (the NOMINATE scale for the per-term and fixed-item scores, chamber standard deviations for the bridged scores), a different metric from the common-space units in which the effect's size is stated. They establish whether a channel operates and in which direction, not its arithmetic share of the adoption effect; the quantitative reconciliation of magnitudes, the vintage arithmetic of the exit-margin subsection, is computed on the common space itself. Appendix A states the scale relation in full.

Table 1: The Time Path of the Polarization Effect: Event-Time Estimates on the Common Space and on the Roll-Call Panel

Years since implementation	Binned TWFE		Sun-Abraham	
	Estimate	<i>p</i>	Estimate	<i>p</i>
<i>Panel A: Common space (Shor-McCarty party-median distance, 1993–2016)</i>				
Lead –7 and earlier	+0.036	.46	+0.006	.80
Lead –6 to –4	–0.026	.41	–0.023	.16
Lead –3 to –1		reference		
Post 0–3	+0.082	.006	+0.079	< .001
Post 4–7	+0.120	.007	+0.118	.01
Post 8–11	+0.183	.01	+0.179	.01
Post 12+	+0.192	.05	+0.168	.07
Years 8–11 minus 0–3	+0.101	.09	+0.099	.08
Years 12+ minus 8–11	+0.009	.86	–0.010	.86
<i>Panel B: Roll-call panel (log party-median distance on fixed-item scores, 1991–2025)</i>				
Lead –7 and earlier	–0.122	.10	–0.186	.08
Lead –6 to –4	–0.031	.30	–0.070	.05
Lead –3 to –1		reference		
Post 0–3	+0.139	.004	+0.140	.04
Post 4–7	+0.124	.14	+0.122	.27
Post 8–11	+0.176	.08	+0.187	.21
Post 12+	+0.131	.28	+0.168	.37
Years 8–11 minus 0–3	+0.037	.57	+0.047	.62
Years 12+ minus 8–11	–0.044	.23	–0.019	.73
Years 20+ minus 12–19	+0.026	.65	+0.076	.42
Chamber-terms after 2016 minus covered 12+	+0.033	.61	+0.032	.63

Note: Panel A reconstructs the Olson and Rogowski (2020) design: Shor-McCarty House party-median distance, 1993–2016, state and year fixed effects, their controls, SE clustered by state; two-way fixed effects +0.128 ($p = .008$), the Sun-Abraham average treatment effect +0.135 ($p = .008$). Panel B: our roll-call panel, 34 chambers (22 term-limited, 12 never-limited), one observation per chamber-biennium with at least five scored members per party (449), 1991–2025; the outcome is the log of the distance between the Republican and Democratic medians of the fixed-item scores; chamber and biennium fixed effects, SE clustered by state. In every column the reference is the three years before implementation (–3 to –1; in Panel B the last pre-implementation biennium) together with never-limited chambers. The far lead (years –7 and earlier) rests on the few units observed that early, one to three states in Panel A and three chambers in Panel B, and is read accordingly. The four post windows are also jointly indistinguishable from one another (binned $p = .38$, Sun-Abraham $p = .14$ in Panel A; .34 and .68 in Panel B). Panel A’s 12+ Sun-Abraham window averages years 12–19, its lead windows are jointly null (binned $p = .19$, Sun-Abraham $p = .30$), and the Sun-Abraham variance matrix is projected to the nearest positive-semi-definite matrix where small cohorts make it indefinite. Roll-call scores carry no common metric across chambers, so Panel B is read for sign and shape only. Further checks: Appendix A, Table A1.

4.2 The Entry Door Is Closed

If term limits polarize by recruiting extremists, entrants in term-limited chambers should be more extreme, relative to the incumbents they join, than entrants in control chambers.

They are not (Table 2). The pooled entrant differential is a precise zero, -0.016 on the W-NOMINATE scale (standard error 0.014), about 1 percent of the 1.3-unit gap between the party means, with differentials above about 0.03 detectable, and the party split points in opposite directions and offsets: Republican entrants in term-limited chambers are *less* extreme than their control-chamber counterparts by 0.056 (standard error 0.022), some 4 percent of the gap between the party means, Democratic entrants slightly more, by 0.021 (standard error 0.011). Nor is the null an artifact of averaging over eras: under chamber-specific event time, the entrant differential is -0.021 inside the transition decade itself (standard error about 0.014), exactly when the compositional shift was underway, and -0.016 in the steady state (standard error about 0.013), neither distinguishable from zero and both leaning negative. The door did not open for more extreme entrants even during the transition.

4.3 The Limit Does Not Move Minds

If term limits polarize by changing members in place, whether because termed-out members are freed to vote their convictions or because careers drift under the compressed horizon, the change should appear within persons, and it should move members outward, toward their own party's pole. It does not. On the measure of Fouirnaies and Hall (2022), the absolute W-NOMINATE score, a member's position in the final permitted term is the same as in her earlier terms: pooled across the 14 term-limited chambers with person and chamber-by-term fixed effects, the final-term shift is -0.001 on a scale whose chamber-term standard deviation is 0.70 and whose party means sit 1.3 apart, and the design detects shifts above about 0.015 (Table 3, Panel A); on the fixed-item and bridged scores it is -0.011 and -0.009 , also not distinguishable from zero. Within party the final term sits a little closer to the center than the member's earlier terms, by about 0.02 for both parties on the W-NOMINATE scale, but that is the late-career accumulation of a drift every member shows, not a response to the cap: once terms served enters the regres-

Table 2: Entry and Exit: Extremity Contrasts Against Control Chambers

	(1)	(2)	(3)	(4)	(5)
		Entrant – incumbent			Exiter – stayer
	All Toward own pole	Republicans Toward own pole	Democrats Toward own pole	All Absolute score	All Toward own pole
Gap in control chambers	0.017 (0.012)	0.058** (0.020)	−0.011 (0.009)	0.016 (0.010)	−0.019* (0.007)
Term-limit differential	−0.016 (0.014)	−0.056* (0.022)	0.021 (0.011)	−0.018 (0.011)	0.006 (0.010)
Chamber × term FE	Yes	Yes	Yes	Yes	Yes
Observations	28,799	15,744	13,054	28,799	27,059
Chambers (clusters)	24	24	24	24	24
Outcome mean	0.60	0.54	0.67	0.62	0.60

Note: Five regressions of member extremity on the entry (columns 1–4) or exit (column 5) contrast and its interaction with term-limit status. Outcome in per-term W-NOMINATE units, the measure of Fourniaies and Hall (2022) (the scale’s chamber-term standard deviation averages 0.70, the gap between the party means 1.3). A positive coefficient means more extreme, a negative one closer to the center. Exiter = last observed term, each chamber’s final panel term excluded as right-censored. Transition-window entry test under chamber-specific event time: term-limit entrant differential −0.021 ($p = .13$) in the transition decade, −0.016 ($p = .22$) in steady state; Republicans −0.061 ($p = .002$) and −0.068 ($p < .001$). Per-chamber entrant gaps average −0.011 (term-limited) versus +0.023 (control). Exiters everywhere sit closer to their party center than stayers (selection relative to the sitting chamber); the term-limit differential is positive but not distinguishable from zero ($p = .56$; the design detects differences above about 0.03), so the mechanical door removes moderates at no higher a rate than electoral churn. Within term-limited chambers, forced (termed-out) Republican exits are disproportionately moderate (−0.031, $p = .002$). Standard errors clustered by chamber in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

sion, the final-term shift is −0.01 for both parties, not distinguishable from zero on any estimator. Chamber by chamber, the shift is null in 26 of 28 party-chamber cells under the two-estimator rule (Appendix Table A3), and the two exceptions run in opposite directions: Maine House Republicans move toward their pole, Michigan House Democrats toward the center. Read on either estimator alone the count is larger (two cells on per-term W-NOMINATE, eight on the fixed-item scores), but the extra cells split between extremizing and moderating and the two estimators agree on none of them. Becoming termed-out does not move a member, toward the poles or away from them; the finding of Fourniaies and Hall (2022) that legislators released from reelection do not change their roll-call positions holds on this panel on three estimators.

Career-long drift tells the same story (Table 3, Panels B and C). Pooled across all

Table 3: Within-Person Position Change Under Term Limits: Pooled Estimates

Outcome: extremity in each estimator's own units (All rows: absolute score; party rows: score toward own-party pole; – = toward the center)					
	Per-term W-NOMINATE	Fixed-item Nokken–Poole	Bridged dynamic IRT	Legislator- terms	Members
<i>Panel A: Final-term shift, term-limited chambers</i>					
All	–0.001 (0.005)	–0.011 (0.010)	–0.009 (0.009)	13,666	4,289
Net of career drift (terms served added)	–0.005 (0.005)	–0.010 (0.009)	–0.015 (0.008)		
Republicans	–0.019 (0.009)	–0.038 (0.019)	–0.025 (0.015)	8,068	2,516
Net of career drift (terms served added)	–0.005 (0.011)	–0.014 (0.015)	–0.012 (0.014)		
Democrats	–0.025 (0.012)	–0.041* (0.019)	–0.031 (0.018)	5,598	1,773
Net of career drift (terms served added)	–0.007 (0.008)	–0.015 (0.019)	–0.015 (0.017)		
<i>Panel B: Career drift per term served, all 24 chambers</i>					
All				26,134	7,329
Terms served, never-limited chambers	–0.006 (0.006)	–0.020 (0.013)	–0.010 (0.010)		
Terms served, term-limited chambers	+0.005 (0.004)	–0.004 (0.008)	+0.006 (0.009)		
Difference (term-limited – never-limited)	+0.011 (0.007)	+0.016 (0.015)	+0.016 (0.013)		
Republicans				14,382	4,067
Terms served, never-limited chambers	–0.026 (0.014)	–0.042** (0.015)	–0.028 (0.014)		
Terms served, term-limited chambers	–0.022 (0.015)	–0.042 (0.023)	–0.023 (0.020)		
Difference (term-limited – never-limited)	+0.003 (0.020)	+0.001 (0.027)	+0.004 (0.024)		
Democrats				11,751	3,262
Terms served, never-limited chambers	–0.023 (0.029)	–0.026 (0.046)	–0.018 (0.036)		
Terms served, term-limited chambers	–0.028* (0.014)	–0.043** (0.014)	–0.027 (0.016)		
Difference (term-limited – never-limited)	–0.005 (0.032)	–0.017 (0.048)	–0.009 (0.039)		
<i>Panel C: Republican-minus-Democratic divergence per term served, all 24 chambers (– = gap narrowing)</i>					
Republican – Democratic difference in the slope on terms served				26,134	7,329
Never-limited chambers	+0.011* (0.004)	+0.015 (0.007)	+0.067*** (0.009)		
Term-limited chambers	+0.004 (0.009)	+0.002 (0.008)	+0.053** (0.014)		
Difference (term-limited – never-limited)	–0.007 (0.010)	–0.013 (0.011)	–0.014 (0.017)		

Note: Outcome: each estimator's score in its own units, following Fourniaies and Hall (2022): per-term W-NOMINATE and fixed-item Nokken–Poole on the NOMINATE scale (Republican-positive; the chamber-term standard deviation of W-NOMINATE averages 0.70 and the gap between the party means 1.3), the bridged dynamic IRT score divided by its chamber's standard deviation across all terms. All rows use the absolute score, party rows the score signed toward the own-party pole, Panel C the signed score. Fixed effects: member-in-chamber and chamber-by-term; standard errors clustered by chamber in parentheses; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Panel A: the coefficient on the final-permitted-term indicator in the 14 term-limited chambers, as is and with terms served added (net of career drift), for both parties pooled (All) and per party. Panel B: the slope on terms served in never-limited and in term-limited chambers (one regression with a slope per regime) and their difference (the coefficient on terms served $\times$ term-limited in the baseline-plus-interaction form of the same regression). Panel C: the Republican-minus-Democratic difference in the slope on terms served, by regime, and its difference across regimes; the bridged scale widens term by term within chambers (Appendix A), so its Panel C slopes are reported but not read. The last two columns give each regression's legislator-terms and its members observed in at least two terms (a member seen once carries no within-person information and drops out); the three estimators use the rows on which all three score a member (the per-term estimator on its full panel of 28,799 rows gives final-term shifts of –0.002 pooled, –0.018 and –0.024 per party). A within-person finding counts only if the first two estimators agree in sign at $p < .05$; the bridge is reported, not adjudicating. Chamber-by-chamber estimates: Appendix Tables A3 and A4.

24 chambers and both parties, a member's extremity does not change with terms served: -0.006 per term in never-limited chambers and $+0.005$ under limits on the W-NOMINATE scale, neither distinguishable from zero, with slopes above about 0.012 per term detectable, under one percent of the party gap, and the same null on the other two estimators. Within party, members drift toward the center of their own caucus by two to three hundredths of the scale per term in both regimes (Republicans by 0.026 per term in never-limited and 0.022 in term-limited chambers, Democrats by 0.023 and 0.028), and the term-limited differential is not distinguishable from zero for either party, every estimate within 0.02 of zero and within 0.004 for Republicans; the design detects Republican differentials above about 0.04 per term, but Democratic ones only above about 0.06, larger than the drift itself, so a Democratic differential of the drift's own size cannot be excluded. The Republican-minus-Democratic gap within careers widens by 0.011 per term in never-limited chambers (standard error 0.004) and by 0.004 under limits (standard error 0.009); the difference between the regimes, -0.007 per term, is not distinguishable from zero, and the design detects differences above about 0.03 per term, about two percent of the party gap. Chamber by chamber, some career slope clears both estimators in 13 of the 24 chambers, 6 of the 14 term-limited against 7 of the 10 never-limited, and the significant slopes run in both directions in both regimes: the party gap diverges in five chambers (two term-limited) and converges in two (one term-limited), and Republicans' own slopes run rightward in four and toward the center in five (Appendix Table A4). What moves inside legislative careers is a member's position relative to her own caucus, a few hundredths of the scale per term toward its center in both regimes; the chamber's extremity does not change with tenure, the movement is no more common under limits than without them, and no estimate, pooled or by chamber, shows positions moving outward under the limit. Conversion is not a term-limits channel.

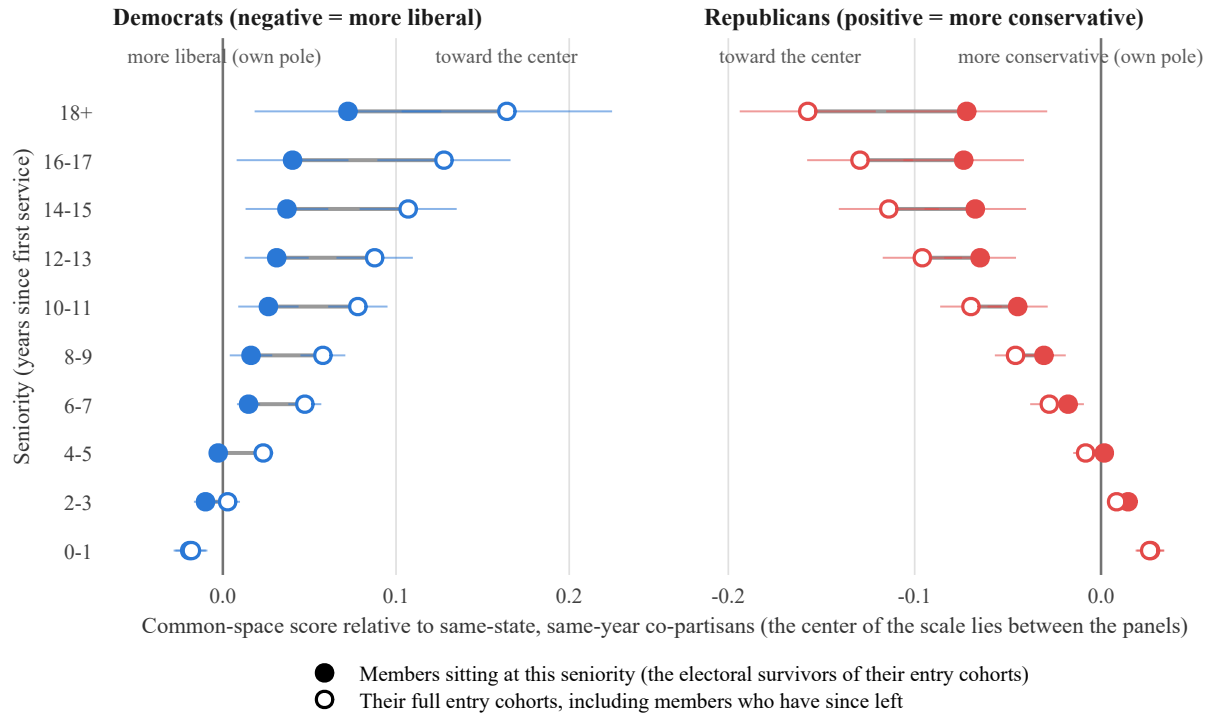
4.4 The Exit Margin: A Door That Opened Once

What remains is exit, and the steady-state evidence closes it from an unexpected direction. Exiting members everywhere sit closer to their party's center than the colleagues they leave behind. Electoral attrition removes the relatively moderate. The term-limit differential on that gap is small and positive, $+0.008$ on the W-NOMINATE scale against an exit gap of -0.021 in control chambers, and not distinguishable from zero (Table 2, column 5; the design detects differences above about 0.02): the mechanical door removes moderates at no higher a rate than electoral churn does in control chambers, which is what a door that cannot see ideology should do when the alternative door can. A steady-state exit channel would require a clearly negative differential.

Why, then, were the members the door first expelled so moderate? By vintage, not by survival. Elections select against moderates *relative to the sitting chamber*; the veteran membership was moderate in *absolute* terms, by virtue of when it had been recruited; and it is that absolute, vintage-based moderation, not the relative selection, that a cap truncates.

The compositional record shows this directly. Figure 1 lays the sitting membership out by seniority on the Shor and McCarty (2011) common space, where scores are career-static and every gradient is therefore compositional: at each layer, the members sitting there against the full entry cohorts they came from. Both move toward the center as seniority rises, the full cohorts faster than the survivors. Entry cohorts drift outward in both parties by roughly a tenth of the scale per decade of entry, and veterans with twelve or more years of service sit center-ward of their most junior colleagues in the same state and year by one-half to one decade's worth of that drift. Electoral attrition works against that gradient: at twelve or more years the survivors sit 0.047 (Republicans) and 0.071 (Democrats) further toward their party's pole on the common-space scale than the full cohorts they entered with, so the gap across cohorts, roughly 0.15 and 0.14, is the larger of the two components. The drift itself differs between term-limited and never-limited states

Figure 1: Recruitment Vintage: The Sitting Chamber as a Layered Archive



Note: Shor and McCarty (2011) individual-level common-space scores, the sample of Appendix Table A2 (49 states, 1993–2020, entrants whose first year of service is observed, 118,431 person-years). Each row is a seniority layer of the sitting membership, years since first service in two-year bins. The horizontal position is a member’s score relative to the mean of the same-state, same-year co-partisans, in each party’s own direction (Democrats: negative is more liberal; Republicans: positive is more conservative), so the center of the scale lies between the two panels. Filled circles: the members sitting at that seniority, the electoral survivors of their entry cohorts. Hollow circles: the full entry cohorts those members came from (same state, party, and entry year), including members who have since left. Because the scores are career-static, the movement across layers is recruitment vintage and the gap within a layer is electoral survival selection; neither can be within-person change. Whiskers: 95 percent confidence intervals, standard errors clustered by state.

by 0.019 per decade for Republicans and 0.024 for Democrats (standard errors 0.036 and 0.035), neither distinguishable from zero, with differences above about 0.07 detectable, below the 0.11 drift itself (Appendix Table A2): the reform does not change what flows in, consistent with the closed entry door; it shortens the chamber’s memory of what flowed in before. A back-of-envelope calculation reconciles the magnitudes. Term-limited membership runs four to five years junior to control membership, which at the observed drift rate translates to roughly 0.10–0.12 on the party-distance scale (the two parties’ outward drifts sum in the distance metric), the same order as the adoption effect.

The door mattered once. Within term-limited chambers, *forced* Republican exits (members expelled by the cap rather than departing voluntarily) are disproportionately moderate, by 0.031 on the W-NOMINATE scale (standard error 0.008), and by era the forced-exit moderation is largest in the years before the transition, partly reflecting anticipatory retirements by the final pre-limits cohorts. Assembled, the pieces give the polarization effect a complete account: implementation expelled the chamber's oldest and most moderate recruitment vintages, especially on the Republican side, shifting the chamber's location within the transition decade; thereafter the entry door recruits entrants no more extreme than elections do, positions do not change within persons, and the mechanical exit door is no harder on moderates than elections would have been. The moderate vintages left once; entry and exit since then match ordinary elections.

5 Effort

5.1 The Final-Term Decline Is Selective

Termed-out legislators sponsor less. Consistent with Fourniaies and Hall (2022), entering the final term reduces sponsorship by 1.0 bills per term (standard error 0.38), from a mean of 14.6, about 7 percent, and the decline is concentrated in substantive legislation (Table 4, columns 1–2). The decline does not reach the law. Bills becoming law show no detectable final-term change, a decline of 0.15 laws on a mean of 4.4 (standard error 0.23); that count alone cannot exclude a fall as large as the 7 percent fall in sponsorship, but the per-bill enactment rate, estimated far more precisely (a Poisson estimate of 0.003, standard error 0.029), does not fall, so termed-out members pass an undiminished share of a smaller portfolio (columns 3–4). The stage profile pins down where the shed effort was going: the decline is largest at introduction, attenuates stage by stage as bills advance (from about 6 percent of the stage mean at introduction to under 2 percent at chamber passage), and a stacked test rejects a flat profile (Appendix Table B2). This is the signature

Table 4: Final-Term Effects Within Legislators, by Outcome

	(1) Bills sponsored	(2) Substantive sponsored	(3) Bills becoming law	(4) Substantive becoming law
Termed out	-1.004* (0.376)	-0.920* (0.375)	-0.150 (0.229)	-0.076 (0.216)
Legislator FE	Yes	Yes	Yes	Yes
Chamber $\times$ term FE	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes
Observations	7,611	7,611	7,611	7,611
Outcome mean	14.6	13.3	4.4	3.9

Note: Sample: term-limit-binding lower chambers, 1993–2017. Controls: majority party, leadership, committee chair, vote share. The per-bill enactment rate is unchanged in the final term (Poisson estimate 0.003, SE 0.029). Standard errors clustered by state. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

selectivity predicts and shirking does not: the bills termed-out members stop writing are disproportionately the ones that would have died early. And the reallocation responds to the horizon that drives it: the sponsorship decline is 1.5 bills in lifetime-limit states, where exit is permanent (standard error 0.55), against 0.6 in consecutive-limit states (standard error 0.34); the smaller estimate is not distinguishable from zero, and that design detects declines above about 0.7 bills, so a consecutive-limit decline of half the lifetime size cannot be excluded (Appendix Table B3).

5.2 Who Compensates, and Through What Channel

The chamber’s output depends on whether anyone absorbs the effort termed-out members shed. Freshmen do, in bills sponsored. A freshman in a term-limited chamber sponsors 4.5 more bills per term than a freshman in a never-limited chamber (standard error 1.8; Table 5, Panel A). The right way to size that effect is against the deficit it erases. In never-limited chambers a freshman sponsors about six and a half fewer bills than a returning colleague; the boost cuts that deficit to about two bills, closing roughly two-thirds of it. Newcomers supply activity rather than statutes: their laws passed differ by 0.06 (standard error 0.39), and that design detects differences only above about 0.8 laws, twice the

female gain reported next, so the freshman result is a sponsorship result; that is what inexperienced compensators should look like, and the mirror image of the termed-out members (inputs down, outcomes flat).

Women compensate in bills that become law, and within their own careers. Women in term-limited chambers pass 0.37 more bills into law per term than women in never-limited chambers, relative to men (standard error 0.15), about 8 percent of the female mean; their sponsorship differs by 2.1 bills (standard error 1.1), not distinguishable from zero (Panel A, columns 1 and 3). Panel B shows the gain is behavioral, not compositional: it survives and roughly doubles with legislator fixed effects, and it grows with years since limits bound, reaching 1.0 additional laws per term (standard error 0.33), a fifth of the female mean, after nine or more years under limits. Selection fails where the tests have power: the share of women serving is 2.0 percentage points lower in term-limited chambers (standard error 1.6), not distinguishable from zero, and the gain survives legislator fixed effects, which selection cannot produce. The cohort test is uninformative: women who entered after limits bound pass 0.55 more laws per term than those who entered before (standard error 0.41), a difference not distinguishable from zero but of the gain's own size, so it neither supports nor rejects selection. At the moderator level emphasized by Holman and Mahoney (2023) we do not replicate their asymmetry (the final-term sponsorship decline is 1.2 bills for women, standard error 0.9, and 1.0 for men, standard error 0.5, and the difference is not distinguishable from zero, though with standard error 0.7 that test detects only differences above about 1.4 bills), but their core claim holds at a margin their design could not isolate: the same women pass more law under term limits, through effort rather than selection.

Two potential channels could produce this compensation. The post channel is concrete: as limits expel senior members, the committee chairs and leadership posts they vacate pass down to juniors, and the gains we measure would then reflect what those posts confer on their new holders. The effort channel is diffuse: the cap shortens ev-

Table 5: Compensation Under Term Limits: Groups, Margins, and Careers

	(1) Bills sponsored	(2) Substantive sponsored	(3) Bills becoming law	(4) Substantive becoming law	(5) Unnormalized SLES
<i>Panel A: Cross-member interactions (state + term FE)</i>					
TL × Freshman	4.506* (1.825)	4.662* (1.832)	0.059 (0.392)	0.190 (0.370)	39.59 (21.23)
TL × Female	2.057 (1.145)	2.055 (1.131)	0.365* (0.152)	0.331* (0.143)	28.11* (13.61)
<i>Panel B: The female gain within individual careers</i>					
TL × Female (legislator FE)			0.681* (0.279)	0.485 (0.309)	27.06 (14.29)
× 9+ years exposure			1.009** (0.332)	0.972** (0.282)	44.46* (21.49)
Observations (Panel A / B)	41,770 / 37,474				

Note: Panel A: each cell is the term-limit interaction from a separate regression with group main effects and individual controls; leader and committee-chair interactions are null on sponsorship and on the SLES (Appendix Table C1). Panel B adds legislator fixed effects (identifying from women observed under both regimes) and, in the second row, interacts the female effect with 9+ years since limits bound. Standard errors clustered by state. ** $p < 0.01$, * $p < 0.05$.

ery member's expected career from the day of enactment, giving newcomers a reason to legislate at once rather than defer to seniority, whether or not any post changes hands.

Leaders and committee chairs show no compensating gain: the term-limit interactions are -8.8 bills for leaders (standard error 6.7) and -2.2 for chairs (standard error 3.4) on sponsorship, and -122 and -0.4 on the unnormalized SLES (standard errors 88 and 49), none distinguishable from zero and every point estimate negative; the leader test is imprecise, detecting differences only above about 13 bills, so a leader gain of the freshman boost's size is excluded by the sign of the estimates and the timing evidence below rather than by the interaction alone (Appendix Table C1). Posts themselves do rotate downward: committee chairs in term-limited chambers average 2.7 terms of service against 6.1 in never-limited chambers, and freshman chairs are four times as common. But the post channel and the effort channel are independent: controlling for one leaves the other unchanged. Controlling for chair and leadership posts changes the freshman boost by under 2 percent; termed-out members keep their chairmanships in the final term, so the posts

redistribute through exit, not mid-career reallocation; and termed-out chairs cut sponsorship by 0.34 bills more than termed-out members without chairmanships (standard error about 1.1), not distinguishable from zero, though that probe detects differences only above about 2.3 bills (Appendix Table C1).

The timing evidence points the same way: the freshman boost does not scale with how many seniors are exiting in a given term (its interaction with the chamber's termed-out share is 1.6 bills, standard error about 6.4, an imprecise test), is equally large under lifetime and consecutive limits (4.2 and 4.7 bills), and, most tellingly, is already at full size in adopting states *after enactment but before the limits had expelled anyone*, remaining flat for two decades thereafter. That profile rules out the alternatives that expulsion triggers: compensation through vacated posts and through the erosion of norms as veteran enforcers exit both require seniors to leave, and the boost was at full size before anyone had left. The evidence therefore points to the effort channel over the post channel. The selection alternative is excluded only in its exit-triggered version, new entrant types arriving to replace the expelled. Selection at enactment is not excluded: candidates who enter after the cap passes already face a bounded career, so a changed entrant pool would produce the same immediate, dose-flat profile as changed behavior by unchanged entrant types. Under both readings the response is to the regime rather than the dose and dates from the day the rules changed; whether the cap changed what entrants do or who the entrants are, these data cannot separate the two.

One caveat applies. The bill histories underlying the SLES begin in 1993, three years after the first limits passed and a year after most of the rest, so the adopting chambers enter our data with their limits already on the books. Of the thirteen treated states, only Nevada, which adopted in 1996, is observed before enactment, and only for a single term; for the other twelve, pre-enactment sponsorship is simply never observed. The immediate full-size boost is therefore identified from the contrast between adopters and non-adopters, not from any within-state change at enactment. That comparison cannot reject

one further alternative: the sponsorship gap between freshmen and returning members may already have been small in the adopting states before limits existed. The concern is specific to how limits were adopted. Limits passed almost exclusively by ballot initiative, and the same anti-incumbent political culture that ran those campaigns could have kept deference to seniority weak in those chambers long before limits arrived. A gap that was never large and a gap erased at enactment look identical in data that begin after enactment, so we read the compensation channel as a conditional association.

5.3 Output Preserved, Distribution Recut

The mechanisms sum to a stable aggregate. Operating under limits does not change measured output: the individual-level unnormalized SLES is 11.4 units higher in term-limited chambers (standard error 20.8), the chamber-term total 640 units lower (standard error 1,151, 3 percent of the mean of 21,547), and the chamber mean 2.7 units lower (standard error 19.3, under 1 percent of the mean of 358), none distinguishable from zero and the chamber-total estimate under a tenth of a chamber-term standard deviation. The chamber-level designs detect changes above about 11 percent of the mean, so the aggregate alone cannot exclude a loss of a few percent; what excludes it is the mechanism evidence, the null on termed-out members' laws and the freshman compensation. What changes is who claims the credit: returning members' median advantage over freshmen in normalized SLES narrows by 0.13 (standard error 0.05), about 40 percent of its mean, while no gap in unnormalized SLES changes detectably (the unnormalized seniority gap falls by 25.9 units, standard error 15.0, not distinguishable from zero at the 5 percent level; Appendix Table B4). Term limits do not shrink the legislative pie; they recut it toward newcomers. One qualification: nine or more years after limits bind, chamber output of substantive law is lower by 58 laws per session (standard error 33), not distinguishable from zero at the 5 percent level, and the design detects declines only above about 66 laws, so the slow capacity erosion Kousser (2005) describes cannot be excluded; the stability we

document is a medium-run fact.

6 Two Clocks, One Legislature

The position result and the effort result are one finding. A legislature's ideological location is embodied in persons whose positions the reform leaves in place, so it moves only when the persons change, and term limits change the persons once, when the limits first bind and the mechanical door releases the moderate vintages that decades of outward-drifting recruitment had deposited. A legislature's effort is rechosen every term under the incentives then in force, and term limits change those incentives for every cohort thereafter: termed-out members prune their portfolios, and juniors' horizons are short from the day they arrive. Once stated, the asymmetry organizes the literature's contradictions. Chamber polarization with stable individual ideology is not a paradox; it is what a compositional adjustment looks like at the two levels of observation. Individual disengagement with stable aggregate output is not a paradox either; it is what a compensated reallocation of effort looks like.

The asymmetry even reappears inside each dimension. On the position side, the one movement we detect is the compositional shift; every member-level channel is closed. On the effort side, the final-term withdrawal (a per-cohort incentive response) is sensitive to the cap's design, its point estimate nearly tripling where limits end careers permanently; the juniors' compensation (a regime-level change in expected career horizons) is uniform across cap designs and at full size from the day the rules changed. Where outcomes are embodied in people, term limits act through composition, once; where behavior is chosen, term limits act through incentives, continuously.

7 Robustness

Three families of checks discipline the design; details are in the appendix. First, the two panels describe the same institutions and the same transition. Re-estimating the position headlines inside the effort panel’s 1993–2017 window preserves every verdict, with the forced-exit moderation, if anything, sharper (from -0.031 to -0.033), and restricting the effort panel’s treated states to the twelve that appear in the roll-call panel moves the termed-out sponsorship decline from 1.00 to 1.04 bills, the freshman boost from 4.5 to 4.4, and the female gain in laws from 0.36 to 0.32, every verdict unchanged (Appendix D); and the chamber-level transition path re-estimated on the roll-call panel’s own scores reproduces the sign and the flat post-implementation shape of the common-space path and extends its plateau to 2025 (Table 1, Panel B), a plateau that holds under alternative member thresholds and clustering and on a vote-direction measure that needs no scaling model (Appendix A). Second, every within-person position claim clears the two-estimator rule described above; single-estimator findings (including several that per-term scaling alone would have certified, and the wider set the bridged scores alone would have) are reported only as such. Third, the cross-chamber effort estimates draw their identifying variation from thirteen treated states, so we subject them to leave-one-state-out estimation and a wild cluster bootstrap (Cameron et al., 2008); the freshman and female estimates hold under both, as does the aggregate null, and the distributional seniority-gap estimate holds at the 5 percent level under the wild cluster bootstrap and in 40 of the 43 leave-one-state-out samples.

8 Conclusion

Term limits were adopted to change who legislates, and they were evaluated by two literatures that reached opposite kinds of verdicts: the reform polarized chambers while its members’ positions stood still, and it left lawmaking output intact while its members dis-

engaged. We have shown that both verdicts are correct and neither paradox is real. Term limits changed the legislature's people once, and its incentives forever. The polarization cost is a fixed transition cost, the one-time release of the accumulated moderate veterans, completed within roughly a decade of implementation, with no ongoing channel through entry, exit, or conversion. The effort change is ongoing but self-compensating: every term's termed-out members abandon the bills that were going nowhere, and every term's newcomers, facing a compressed horizon from the start, supply the offsetting activity, with women passing more law within their own careers.

The policy lesson follows from the composition–incentive distinction. Debates over term limits price the reform as if its costs ran forever: chambers that keep polarizing, output that keeps eroding. On the polarization side that pricing is wrong, since the effect is real and bounded, while on the output side the feared cost never materializes at all, though the long-run erosion in substantive lawmaking deserves monitoring as post-2017 data accumulate. The lasting change term limits make is subtler than either fear: they flatten the seniority structure of legislative work, moving effort and success from veterans toward newcomers and, in law passed, toward women. Whether that egalitarian recuiting is worth the documented costs in expertise and oversight (Kousser, 2005; Carey et al., 2000, 2006) is a normative question, but the ledger should carry the right entries.

The analysis has limits. Term-limit adoption is endogenous: the cross-chamber effort comparisons and the regime-level reading of the compensation channel remain conditional associations, and a pre-existing seniority culture in adopting states cannot be fully excluded, though the within-career designs that support the causal claims are immune to adoption timing. Roll-call positions do not exhaust representation, and bill counts do not measure oversight or quality. Committee records, which would make agenda control directly observable, and candidate-level linkages tracing where termed-out members go next are the natural next steps.

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A Roll-Call Measurement and the Full Census

The roll-call pipeline scores every chamber-term with W-NOMINATE from the state's own recorded votes, orients scores with party labels cross-validated against Shor and McCarty (2011), and links members across terms through a person-level identity cross-walk. Party separation (the correlation between score and party) ranges from 0.73 to 0.98 across the 34 chambers. Because per-term scaling can stretch or compress the score space differentially across terms (manufacturing spurious within-person movement), every within-person claim in the paper is required to clear two estimators whose failures run in opposite directions, and a third is reported beside them. Per-term W-NOMINATE rescales every term independently, so a term-specific stretch of the score space registers as movement of every member. The fixed-item Nokken–Poole-style estimator (Nokken and Poole, 2004) estimates the items once, from a pooled scaling in which each member holds a single position, and then frees each member-term against the pinned items; a movement common to a whole party is partly absorbed into the items in the first step, a member's movement relative to the chamber is not. The bridged dynamic item-response (IRT) model (Imai et al., 2016) ties each member's positions across terms through a random-walk prior (variance 0.10 per biennium, 0.50 as a sensitivity) while estimating the items term by term, so nothing but that prior and the entrants' initial-position prior connects the scale of one term to the next. The identification limit is common to all three: within a term, multiplying every member's position by a constant and dividing every item's discrimination by the same constant leaves the likelihood unchanged, so a chamber's roll calls identify a member's movement relative to the chamber-term, not the chamber's own movement, which is why the chamber-level effect is estimated separately (Section 3.2). The two adjudicating estimators measure that relative movement; the bridge lets the chamber's scale change across terms as well, and its priors decide how the two are split. Every contrast between members that crosses chamber-terms standardizes the score within chamber-term first.

The full census of the 24 member-level chambers (replication archive) records how the three behave. On the Republican-minus-Democratic differential in the career slope, per-term W-NOMINATE returns a significant divergence in 10 chambers and a significant convergence in 4; the fixed-item estimator, 5 and 2; the bridge, 15 and none. The bridged scale widens across terms: the distance between the bridged party medians grows from first term to last in every one of the 24 chambers, by a median 5 percent per biennium, more than twice the growth of the common-space distance in the same chambers over the overlapping years (2 percent) and of the fixed-item distance (2 percent), and continuing members' bridged scores move apart by party at about two-thirds of that rate, so most of the widening is movement of continuing members, not turnover. It is not a start-of-career artifact: it is linear across the career and survives dropping every member's first observed term. Whether that widening is polarization within persons that the other two estimators cannot see or a consequence of tying the scale across terms by prior alone cannot be settled on the roll calls of one chamber; what can be said is that the bridge returns a divergence within persons far more often than the other two and a convergence nowhere, in both regimes alike (7 of the 14 term-limited chambers, 8 of the 10 never-limited). It therefore does not adjudicate. Requiring the per-term and fixed-item estimators to agree admits both directions (five diverging and two converging chambers on the differential), and the bridge agrees with every diverging verdict but one and with neither converging one. On the 28 termed-out cells the same rule leaves 26 null; per-term W-NOMINATE alone flags two (one extremizing, one moderating), the fixed-item scores alone eight (two and six), and the bridge seven (two and five). Table A3 prints every cell and marks the two that clear the rule.

The scale relation between the two panels deserves explicit statement. The adoption effect is estimated in Shor and McCarty (2011) common-space party-distance units, across all states, 1993–2016; the member-level channel tests are estimated in the roll-call scores' own units on the roll-call panel's 24 chambers without staggered terms, 1991–2025, per-

term W-NOMINATE and the fixed-item scores on the NOMINATE scale and the bridged scores in chamber standard deviations, the measure Fourniaies and Hall (2022) use for the same question. Per-term W-NOMINATE carries no common metric across chambers or terms beyond the unit-circle normalization of each scaling, so member-versus-member contrasts are well defined only within a chamber-term, which the chamber-by-term fixed effects of every member-level design enforce and which blocks the cross-scale level comparisons that would otherwise be read as ideological movement. An earlier version of these tests standardized each score within its chamber-term (mean 0, SD 1). That standardization is not innocuous: the chamber-term standard deviation of per-term W-NOMINATE ranges from 0.28 to 0.93 across the panel, so standardizing re-weights chamber-terms by the inverse of their dispersion and turns growth in a chamber's dispersion into apparent movement of its continuing members toward the center. Every verdict in Tables 2 and 3 is the same on either metric except two, both recorded here: on standardized scores the within-career Republican-minus-Democratic gap narrows by 0.02 to 0.03 standard deviations per term in never-limited chambers, where on the scores' own units it widens by 0.011 (standard error 0.004), and the term-limit differential on the exit gap is +0.026 standard deviations (standard error about 0.013) where Table 2 reports +0.006 (standard error 0.010); neither changes what the tests establish, the presence, sign, and stability of member-level channels in the steady state, not a decomposition of the adoption effect into channel shares. The price of the own-units metric is unchanged: no unit conversion exists between it and the common space. The quantitative reconciliation the paper offers accordingly never crosses scales: the vintage gradients and seniority-gap arithmetic of Table A2 are computed on the common space, in the same units as the adoption effect, and the window mismatch is checked separately (every position headline holds inside the 1993–2017 effort-panel window; Appendix D). Because five of the 14 term-limited chambers enter the roll-call panel only after their limits bound, the pooled channel tests speak mainly to the steady state; the transition-window entry test draws

on the chambers observed before binding, and the chamber-level transition path is documented on the common space and on the roll-call panel's own fixed-item scores (Table 1, Panels A and B); Table A1 reports the roll-call path under alternative member thresholds and clustering and on a vote-direction measure that needs no scaling model.

Table A1 holds the checks on the roll-call path. On the five-member panel of Table 1, Panel B, the post windows are jointly distinguishable from zero and cannot be told apart from one another; the rise across the first decade is $+0.037$ (standard error about 0.07) and the change after year twelve -0.044 (standard error about 0.04). The common-space path in Panel A has the same profile: the rise across the first decade is $+0.101$ (standard error about 0.06), the change after year twelve $+0.009$, and a one-sided test of growth continuing at the transition rate does not reject at conventional levels. In both columns of Panel B the reference is the last pre-implementation biennium, because only three chambers (both Nevada chambers and the Arizona House) are observed seven or more years before implementation; on these scores that biennium already sits above the far lead, so the post windows are measured from a biennium in which the gap may have begun to rise. Raising the member threshold from five to eight per party drops the chambers with small minority caucuses and lowers every post-window estimate by a tenth to a third without moving the sign or the flat shape; clustering by chamber instead of state leaves the estimates unchanged and the inference similar; and neither a three-member threshold nor dropping New Jersey (whose even-year terms are mapped to the next odd year) changes any reading. The vote-direction measure replaces the scaling model with a count: each member's share of votes cast on the conservative side of the roll call, the side being defined either by the sign of the item's discrimination parameter in the fixed-item model (item-sign rule) or by the side the Republican majority took on roll calls where the two party majorities opposed each other (majority rule, which is circular with respect to party discipline), summarized as the gap between the two parties' median shares. Under the item-sign rule the gap in term-limited chambers was already well above its far-lead level

Table A1: The Roll-Call Transition Path: Member Threshold, Clustering, and Vote-Direction Checks

Years since implementation	Fixed-item gap				Vote-direction gap			
	≥ 8 per party		Chamber-clustered		Item-sign rule		Majority rule	
	Est.	p	Est.	p	Est.	p	Est.	p
Lead -7 and earlier	-0.079	.30	-0.122	.21	-0.407	.008	-0.042	.47
Lead -6 to -4	-0.004	.89	-0.031	.51	-0.022	.73	+0.010	.70
Lead -3 to -1				reference				
Post 0-3	+0.118	< .001	+0.139	.005	+0.121	.03	+0.077	.01
Post 4-7	+0.110	.03	+0.124	.09	-0.043	.80	+0.054	.10
Post 8-11	+0.113	.10	+0.176	.04	+0.149	.31	+0.060	.21
Post 12+	+0.093	.29	+0.131	.20	+0.053	.74	+0.053	.33
H_0 : four post windows all zero (p)	.004		.01		.04		.006	
H_0 : four post windows equal (p)	.85		.33		.34		.72	
Years 20+ minus 12-19	+0.051	.24	+0.026	.63	+0.062	.41	-0.001	.96
After-2016 cells minus covered 12+	+0.078	.12	+0.033	.57	+0.045	.66	+0.003	.92

Note: Same design as Table 1, Panel B (chamber-by-biennium panel, 34 chambers, 1991–2025, chamber and biennium fixed effects, never-limited chambers and the -3 to -1 window (the last pre-implementation biennium) as reference; estimates in log points of the party-median gap). Column 1: fixed-item scores, chamber-terms with at least eight scored members per party (414; the five-member panel has 449), SE clustered by state (18). Column 2: the five-member panel with SE clustered by chamber (34). A three-member threshold (458 chamber-terms) gives post windows jointly zero $p = .002$, equal $p = .35$, tail $+0.009$ ($p = .88$); dropping New Jersey gives $p = .004$, $p = .34$, and $+0.027$ ($p = .67$). Columns 3–4: gap between the party medians of members’ conservative-vote shares, the conservative side of each roll call defined by the sign of its fixed-item discrimination parameter (column 3) or by the side of the Republican majority on roll calls where the party majorities opposed each other (column 4; on those roll calls the Republican median share exceeds .9 and the Democratic median falls below .1). Under the item-sign rule the far lead sits well below the last pre-implementation biennium (-0.41 , $p = .008$), so the two lead windows are jointly non-null ($p = .03$); relative to that biennium the first post window is $+0.12$ ($p = .03$) and the four post windows are jointly $p = .04$. Only three chambers are observed at years ≤ -7 ; seven of the 22 term-limited chambers enter after implementation. Roll-call scores carry no common metric across chambers, so no column speaks to the size of the common-space estimate.

in the biennium before implementation, so relative to that biennium the first post window rises and the later windows neither grow nor differ from one another, and only the plateau (no growth after onset, no change beyond 2016) is reproduced; under the majority rule the common-space shape appears (flat leads, a positive first post window, flat thereafter) but the party medians lie near the ends of the unit interval and the later windows are imprecise. Both rules agree with the fixed-item path on the absence of growth through 2025; neither independently dates the onset.

Two data-quality notes. First, an identity-resolution audit of the Missouri roster found

Table A2: Recruitment Vintage: Compositional Gradients on the Common Space

Quantity (directional extremity, common space)	Republicans		Democrats	
	<i>b</i>	<i>p</i>	<i>b</i>	<i>p</i>
Entry-cohort drift, per decade of entry	+0.111	< .001	+0.129	< .001
Veterans (12+ years) vs. juniors (0–3 years), same state-year	−0.104	< .001	−0.068	.001
Term-limit differential in entry drift	+0.019	.61	+0.024	.49

Note: Shor and McCarty (2011) individual-level common-space scores, 49 states, 1993–2020, 25,514 legislators; scores are career-static, so every gradient is compositional. Extremity signed toward the own-party pole. Row 1: clean entrants only (states' first two covered years excluded; 9,548 R / 8,033 D), state fixed effects. Row 2: state-by-year fixed effects, persons whose entire career falls inside the covered years (63,556 R / 55,022 D person-years); intermediate tenure bins are monotone. Electoral attrition removes the relatively moderate faster (Table 2), biasing veteran bins toward extremity, so the center-ward gradient is conservative; including censored persons preserves both signs. Row 3: interaction of entry-decade drift with term-limit-state status. SE clustered by state.

same-surname legislators merged into single careers; the repair (splitting 60 merged identifiers into 124 people) eliminated what had appeared to be the panel's strongest termed-out moderation result, and all Missouri figures in this paper are post-repair. The episode motivated checks that each identifier's span of service is a plausible single career, now applied to every chamber's roster. Second, Florida Senate final-term coding approximates the chamber's staggered post-redistricting term calendar; its one significant census cell, Florida Senate Democrats on the per-term estimator alone (Table A3), carries that caveat.

The vintage evidence in the exit-margin subsection comes from the Shor and McCarty (2011) individual-level release rather than the roll-call panel: because its scores are career-static, every tenure or cohort gradient in it is purely compositional, which is exactly the property the vintage claims require. Table A2 reports the three gradients.

Tables A3 and A4 print the chamber-by-chamber census behind Table 3: every termed-out cell on all three estimators, and every chamber's career-drift slopes on the two adjudicating estimators. Per-chamber entry and exit gaps are in the replication archive.

Table A3: Termed-Out Shifts, All 28 Party-Chamber Cells

Chamber	Party	Per-term W-NOMINATE	Fixed-item Nokken-Poole	Bridged dynamic IRT	<i>N</i>	Exception
Arkansas House	R	+0.007 (.93)	+0.014 (.89)	+0.013 (.91)	528	
Arkansas House	D	+0.047 (.53)	+0.045 (.74)	+0.065 (.65)	240	
Arizona House	R	−0.034 (.29)	−0.037 (.41)	−0.024 (.20)	523	
Arizona House	D	−0.014 (.81)	−0.017 (.78)	−0.023 (.48)	303	
California House	R	+0.014 (.46)	+0.027 (.49)	−0.002 (.97)	374	
California House	D	−0.009 (.58)	−0.119 (< .001)	−0.036 (.23)	717	
Florida House	R	−0.038 (.10)	−0.105 (.05)	−0.050 (.17)	1017	
Florida House	D	−0.087 (.18)	−0.087 (.40)	−0.110 (.35)	530	
Maine House	R	+0.044 (< .001)	+0.079 (.007)	+0.134 (.003)	527	yes
Maine House	D	−0.027 (.09)	−0.064 (.01)	−0.086 (.02)	686	
Maine Senate	R	+0.064 (.12)	+0.059 (.34)	+0.072 (.25)	146	
Maine Senate	D	−0.008 (.84)	+0.018 (.77)	−0.011 (.84)	172	
Michigan House	R	−0.018 (.27)	−0.093 (.001)	−0.087 (.008)	671	
Michigan House	D	−0.035 (.02)	−0.055 (.05)	−0.103 (.002)	635	yes
Missouri House	R	−0.001 (.93)	−0.029 (.36)	−0.055 (.02)	1151	
Missouri House	D	+0.014 (.44)	+0.048 (.13)	+0.060 (.07)	664	
Montana House	R	−0.016 (.39)	−0.020 (.42)	−0.033 (.17)	770	
Montana House	D	+0.001 (.92)	+0.010 (.69)	+0.016 (.55)	538	
Nevada House	R	−0.142 (.31)	−0.030 (.84)	−0.099 (.49)	153	
Nevada House	D	−0.013 (.54)	−0.016 (.77)	−0.022 (.34)	233	
Ohio House	R	−0.031 (.27)	−0.067 (.18)	−0.067 (.11)	496	
Ohio House	D	−0.014 (.63)	−0.010 (.88)	−0.089 (.02)	312	
Oklahoma House	R	−0.058 (.28)	−0.055 (.40)	−0.073 (.22)	801	
Oklahoma House	D	−0.101 (.35)	−0.115 (.50)	−0.119 (.44)	418	
South Dakota House	R	−0.009 (.73)	+0.001 (.99)	+0.021 (.26)	662	
South Dakota House	D	−0.021 (.54)	−0.077 (.22)	−0.028 (.22)	189	
South Dakota Senate	R	−0.106 (.06)	−0.140 (.02)	−0.025 (.21)	303	
South Dakota Senate	D	+0.114 (.07)	+0.287 (.003)	+0.129 (.005)	90	

Note: Each cell is the within-person coefficient on the final-permitted-term indicator (person and term fixed effects, SE clustered by person), per party and per chamber, signed toward the own-party pole, with its *p*-value in parentheses; per-term W-NOMINATE on all of the chamber's Republican and Democratic legislator-terms (*N* = its legislator-terms), the fixed-item and bridged estimators on the rows all three score. An exception requires per-term W-NOMINATE and the fixed-item estimator to agree in sign at *p* < .05; the bridge is reported, not adjudicating.

Table A4: Career Drift, All 24 Chambers

Chamber	Regime	Per-term W-NOMINATE			Fixed-item Nokken-Poole		
		R–D	R slope	D slope	R–D	R slope	D slope
Arkansas House	TL	+0.018 (.14)	+0.029 (< .001)	+0.010 (.21)	–0.017 (.41)	+0.006 (.56)	+0.028 (.12)
Arizona House	TL	+0.008 (.72)	+0.007 (.58)	–0.012 (.51)	–0.034 (.20)	+0.013 (.57)	+0.045 (.010)
California House	TL	+0.033 (.010)	+0.009 (.35)	–0.024 (.02)	+0.069 (.02)	+0.031 (.17)	–0.030 (.11)
Florida House	TL	+0.033 (.05)	+0.015 (.03)	–0.014 (.29)	+0.023 (.43)	+0.008 (.60)	+0.002 (.95)
Maine House	TL	+0.014 (.03)	+0.016 (.003)	+0.008 (.25)	+0.009 (.44)	+0.009 (.37)	+0.001 (.89)
Maine Senate	TL	–0.009 (.58)	–0.009 (.49)	+0.003 (.75)	+0.034 (.14)	+0.014 (.51)	–0.009 (.54)
Michigan House	TL	–0.030 (.003)	–0.023 (.03)	+0.015 (.07)	–0.031 (.06)	–0.022 (.04)	+0.010 (.42)
Missouri House	TL	–0.051 (< .001)	–0.029 (< .001)	+0.026 (.01)	–0.039 (.008)	–0.036 (< .001)	+0.015 (.26)
Montana House	TL	+0.035 (< .001)	+0.031 (< .001)	–0.005 (.24)	+0.037 (< .001)	+0.036 (< .001)	+0.001 (.93)
Nevada House	TL	–0.017 (.40)	–0.038 (.10)	–0.038 (< .001)	–0.006 (.80)	–0.017 (.43)	–0.027 (.01)
Ohio House	TL	–0.010 (.57)	+0.007 (.61)	+0.025 (.04)	+0.002 (.96)	+0.007 (.70)	+0.003 (.89)
Oklahoma House	TL	+0.014 (.57)	+0.034 (.005)	+0.002 (.91)	+0.022 (.55)	+0.005 (.73)	–0.023 (.45)
South Dakota House	TL	–0.054 (.01)	–0.080 (< .001)	–0.023 (.08)	–0.030 (.29)	–0.097 (< .001)	–0.065 (.01)
South Dakota Senate	TL	–0.000 (.99)	+0.010 (.78)	–0.006 (.80)	–0.053 (.26)	+0.026 (.26)	+0.052 (.27)
Alaska House	never	–0.048 (.04)	–0.032 (.01)	+0.009 (.67)	–0.082 (.01)	–0.044 (.010)	+0.034 (.19)
Connecticut House	never	–0.002 (.81)	–0.020 (.005)	–0.018 (< .001)	–0.008 (.44)	–0.019 (.02)	–0.012 (.07)
Connecticut Senate	never	+0.018 (.003)	+0.012 (.009)	–0.003 (.43)	+0.021 (.23)	+0.031 (< .001)	+0.010 (.48)
North Carolina House	never	+0.018 (.19)	+0.048 (< .001)	+0.027 (< .001)	+0.033 (.21)	+0.029 (.14)	–0.006 (.70)
North Carolina Senate	never	+0.028 (.01)	+0.034 (.001)	+0.001 (.86)	+0.022 (.33)	+0.059 (< .001)	+0.027 (.09)
New Hampshire House	never	+0.017 (< .001)	–0.006 (.007)	–0.024 (< .001)	+0.018 (.004)	+0.005 (.19)	–0.015 (.003)
New Hampshire Senate	never	+0.024 (.02)	+0.015 (.03)	–0.012 (.009)	+0.055 (.01)	+0.032 (.04)	–0.039 (.001)
New Jersey House	never	+0.017 (.02)	–0.009 (.20)	–0.018 (< .001)	+0.078 (< .001)	+0.019 (.35)	–0.053 (< .001)
Vermont House	never	–0.018 (.17)	+0.024 (.02)	+0.060 (< .001)	–0.017 (.58)	–0.018 (.33)	–0.003 (.90)
Vermont Senate	never	+0.016 (.54)	–0.058 (.005)	–0.065 (< .001)	+0.021 (.48)	–0.020 (.61)	–0.013 (.48)

Note: Slopes on terms served, chamber by chamber, with p -values in parentheses. The R–D differential is the coefficient on terms served $\times$ Republican with person and term fixed effects; each party's slope is the coefficient on terms served with person fixed effects, on the Republican-positive orientation (a negative Republican slope is moderation; a negative Democratic slope is movement toward the Democratic pole). Per-term W-NOMINATE on all of the chamber's Republican and Democratic legislator-terms; the fixed-item estimator on the rows all three estimators score. A chamber clears a criterion in the text's counts only if both estimators agree in sign at $p < .05$.

B Summary Statistics and Additional Effort Results

Table B1: Summary Statistics

	Mean	SD	Min	Max
<i>Panel A: Term-limited lower chambers, effort panel (N = 9,166)</i>				
Bills sponsored	14.6	15.4	0	253
Substantive bills sponsored	13.3	14.1	0	244
Bills becoming law	4.4	5.9	0	89
Termed out	0.18	0.38	0	1
<i>Panel B: All lower chambers, effort panel (N = 41,770)</i>				
Bills sponsored	13.5	16.8	0	393
Bills becoming law	4.5	7.2	0	295
Unnormalized SLES	235	218	0	3,970
Term limit (binding)	0.22	0.41	0	1
Freshman	0.24	0.43	0	1
Female	0.24	0.42	0	1
<i>Panel C: Roll-call panel (N = 33,258 legislator-terms, 34 chambers)</i>				
Term-limited chambers			22 (12 states)	
Never-limited control chambers			12 (6 states)	
Panel span			1991–2025	

Note: Effort panel: Louisiana and Nebraska excluded. Panel B is the cross-member estimation sample: lower-chamber legislator-terms with nonmissing controls, excluding final-term observations. Roll-call panel: per-term W-NOMINATE scored on the project's own pipeline; seven of 22 term-limited chambers enter after limits bound. Member-level contrasts use the 24 chambers without staggered terms, 28,799 legislator-terms.

Table B2: The Final-Term Decline Attenuates Through the Legislative Pipeline

Stage	Outcome	Final-term effect (% of stage mean)
1	Bills introduced	−6.3%
2	Committee action	−4.8%
3	Action beyond committee	−3.4%
4	Chamber passage	−1.7%
5	Becoming law	−3.9%

Note: Stage-specific within-legislator estimates (legislator + chamber-term FE), rescaled by each stage's mean. Stacked test of equal effects at stages 4–5 versus stage 1: Wald $F = 6.85$, $p = 0.001$. Stage 5 has the smallest denominator and is estimated least precisely.

Table B3: Final-Term Effects by Term-Limit Type

	Lifetime limits	Consecutive limits
Bills sponsored	−1.517* (0.545)	−0.568 (0.337)
Substantive sponsored	−1.416* (0.515)	−0.500 (0.366)
Bills becoming law	−0.242 (0.346)	−0.060 (0.286)
Observations	3,130	4,481

Note: Within-legislator specification of Table 4, by limit type. Lifetime: AR, CA, MI, MO, NV, OK. Consecutive: AZ, CO, FL, ME, MT, OH, SD. Standard errors clustered by state. * $p < 0.05$.

Table B4: Term Limits and the Distribution of Effectiveness: Output Versus Shares

	Partisan gap	Chair gap	Seniority gap	Gini
<i>Panel A: Unnormalized SLES (absolute output)</i>				
Term limit	21.93 (15.51)	−4.07 (20.26)	−25.87 (14.99)	−0.020 (0.014)
<i>Panel B: Normalized SLES (within-chamber shares)</i>				
Term limit	0.019 (0.047)	0.044 (0.066)	−0.132* (0.054)	−0.015 (0.013)
Observations	684	692	689	692

Note: Chamber-term level: 684–692 chamber-terms in the 43 states with complete chamber covariates, thirteen of them term-limited; every column conditions on concurrent reforms (redistricting commissions, primary openness, the top-two primary, professionalism) and chamber controls (polarization, seat-share imbalance, log chamber size); state-chamber + year FE; gaps are median group differences. The normalized seniority gap remains detectable when computed from means (−0.097, $p = .045$); the unnormalized gap is marginal ($p = .088$). SE clustered by state-chamber. * $p < 0.05$.

C The Compensation Channel Is Diffuse: Mechanism Probes

Table C1: Post-Holding and Timing Probes of the Compensation Channel

Probe	Estimate	<i>p</i>
<i>Posts rotate downward (allocation, DiD vs. non-TL chambers)</i>		
Chair premium at seniority 2	+0.079	.03
Chair premium at seniority 6+	−0.177	.003
<i>But posts are not the channel</i>		
Gavel retention entering the final term (within person)	+0.093	.26
Termed-out × chair on sponsorship (person FE)	−0.34	.77
Leader × term limit on sponsorship (cross-member)	−8.82	.19
Chair × term limit on sponsorship (cross-member)	−2.22	.51
Leader × term limit on unnormalized SLES (cross-member)	−122.4	.17
Chair × term limit on unnormalized SLES (cross-member)	−0.4	.99
Freshman boost: share mediated by chair/leader posts	1.7%	
Committee throughput vs. termed-out-chair share	−0.007	.86
<i>Timing: the freshman boost is a regime constant (vs. never-TL freshmen)</i>		
Enacted, not yet binding	+4.95	.03
Years 0–3 after binding	+4.54	.04
Years 4–7	+4.89	.02
Years 8+	+4.65	.02
<i>And does not scale with dose or cap design</i>		
Freshman × chamber termed-out share	+1.59	.80
Lifetime-limit states	+4.22	.02
Consecutive-limit states	+4.69	.03

Note: Sources: the replication archive’s mechanism scripts. Chairs in term-limited chambers average 2.7 terms of service versus 6.1 elsewhere, and freshman chairs are four times as common. Termed-out members nonetheless keep their chairmanships (the posts redistribute through exit), termed-out chairs withdraw like other termed-out members, and controlling for posts moves the freshman boost by under 2%. The timing profile (full-size before limits expelled anyone, flat for two decades, dose-flat, uniform across cap designs) is consistent with a career horizon shortened at enactment; because the panel begins after nearly every enactment, the possibility that the gap between freshmen and returning members was already small in adopter states before limits cannot be excluded. Selection of differently ambitious entrants at enactment shares this timing profile and is excluded only in its exit-triggered version (new entrant types replacing the expelled). The freshman additions match the final-term cuts in composition (substantive, early-dying; law counts null on both sides).

D The Two Panels Describe the Same Institutions

Table D1: Effort Headlines on the Roll-Call-Panel States

	Full panel	Treated $\cap$ roll-call states	Strict (never-limited controls only)
Termed-out sponsorship	-1.00*	-1.04*	= col. 2
Termed-out laws (null)	-0.15	-0.08	= col. 2
Seniority gap (norm.)	-0.132*	-0.133*	-0.167*
TL $\times$ Freshman	+4.51*	+4.37*	+0.97
TL $\times$ Female (laws)	+0.36*	+0.32*	+0.15
Aggregate (null)	+11.4	+10.9	+9.5

Note: Column 2 drops the treated states absent from the roll-call panel (CO; LA is absent from the effort panel); every headline is unchanged. Column 3 additionally restricts controls to the six never-limited roll-call states (a stress test, not a design claim): New Hampshire's 400-member chamber is 45% of that control pool and compresses the baseline freshman deficit (from -6.5 to -3.5 bills); dropping NH recovers the freshman point estimate to +2.6 (standard error about 2.3, with five control states). The chamber-level seniority-gap spec, where NH is one chamber rather than 45% of members, is stable throughout.

Table D2: Position Headlines Inside the Effort Panel's Window (1993–2017)

	Full panel (1991–2025)	≤ 2017
Exit differential (exiter $\times$ TL)	+0.006	-0.003
Forced Republican exits (moderation)	-0.031**	-0.033**
Entry differential, transition window	-0.021	-0.023
Entry differential, steady state	-0.016	-0.022
Termed-out census	26 of 28 null	27 of 28 null

Note: Estimates in per-term W-NOMINATE units, as in Table 2. Every verdict is unchanged inside the SLES window: the exit differential is null in both windows, the moderation of forced Republican exits if anything sharper, and the entry differentials null in both windows. The two-estimator census gives 26 of 28 null cells on the full panel and 27 of 28 inside the window: Michigan House Democrats leave the exception list in the shorter window and Maine House Republicans stay, further evidence that the exceptions are noise rather than mechanism.